Torchvision environment setup

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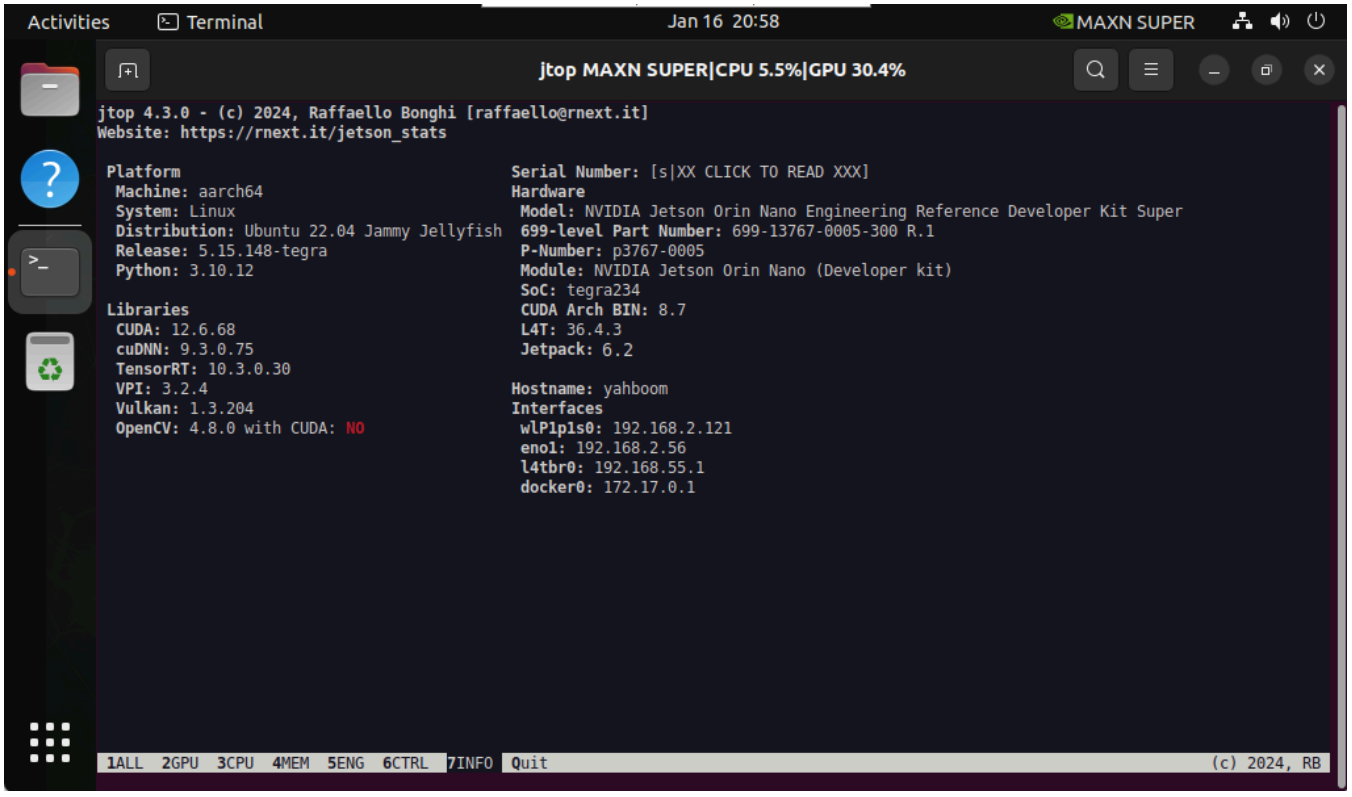
References

> **pip installation note:** Under PEP 668, Ubuntu 24.04 marks the system Python as "externally managed", so installing directly with pip may report `externally-managed-environment`. `--break-system-packages` allows pip to install packages in that environment; virtual environments usually do not need this parameter, but it is kept in this tutorial so that the commands stay consistent.

> The factory system already has the relevant environment set up, so the installation steps can be skipped.

Torchvision provides computer vision datasets, models and image transformations for PyTorch.

1. System information



The current environment uses PyTorch `2.8.0`, Torchvision `0.23.0` and Python `3.12`. Torchvision `0.23.0` corresponds to PyTorch `2.8.0`.

2. Install the dependencies

```
sudo apt update
sudo apt install ninja-build libwebp-dev libjpeg-dev -y
```

3. Install Torchvision

According to the version correspondence between PyTorch and Torchvision, PyTorch `2.8` uses Torchvision `0.23`, which supports Python `3.9` to `3.13`. When installing, use a version that fits Jetson ARM64, Python 3.12 and the current PyTorch.

3.1. Offline installation

Prepare in advance the Torchvision `0.23.0` wheel file that matches the current environment, and copy it to the Jetson. Do not use the `0.20.0a0` / `cp310` wheel from the old tutorial.

```
cd Downloads/
python3 -m pip install --user --break-system-packages ./<Torchvision-0.23.0 ARM64 wheel file name>
```

3.2. Online installation

Install from the PyTorch Vision `v0.23.0` source on a networked device. This method uses the currently installed PyTorch and does not automatically replace its dependencies:

```
python3 -m pip install --user --break-system-packages --no-deps --no-build-isolation "git+https://github.com/pytorch/vision.git@v0.23.0"
```

3.3. Source code compilation

Download the `v0.23.0` source:

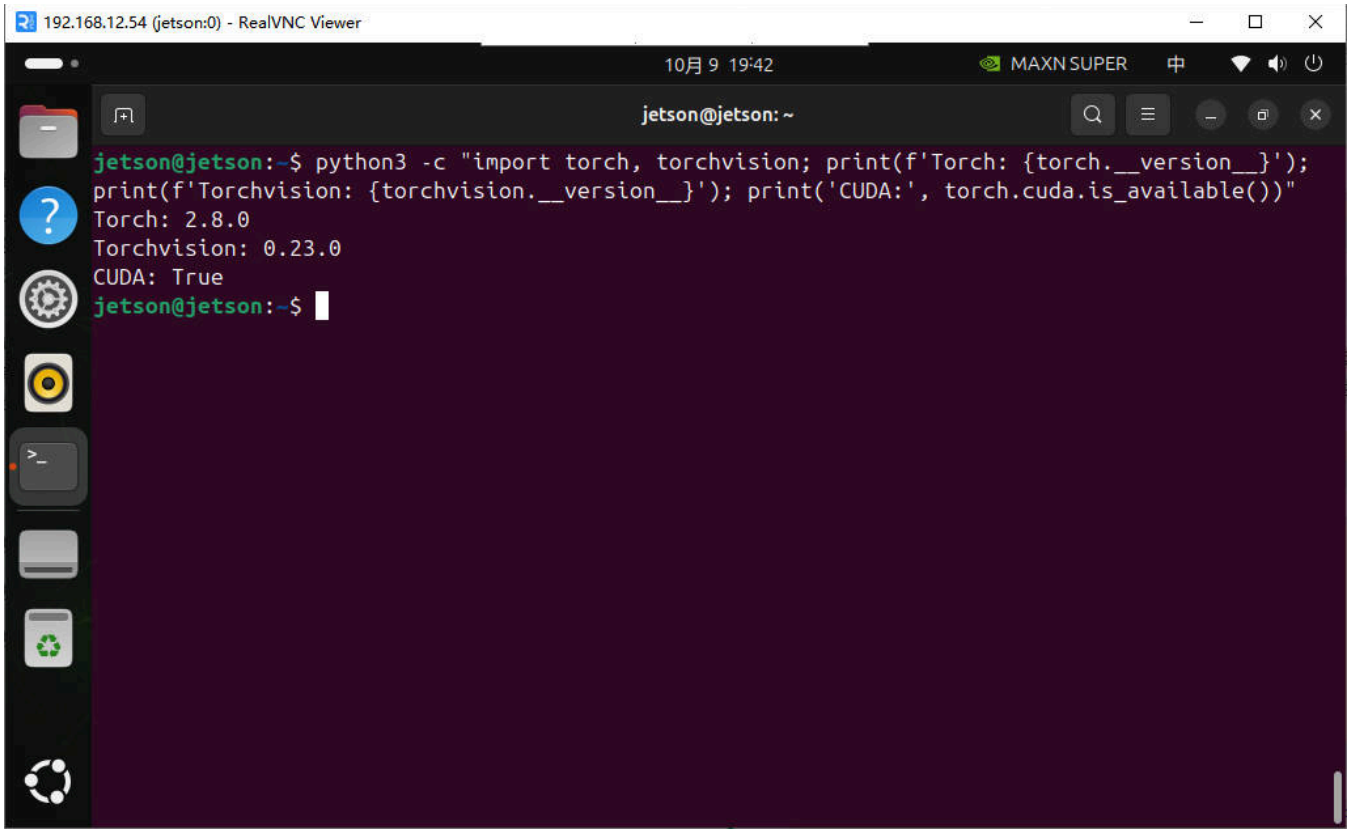
https://github.com/pytorch/vision/tree/v0.23.0

Enter the source directory and then install:

```
cd vision-0.23.0
python3 -m pip install --user --break-system-packages --no-deps --no-build-isolation .
```

4. Verify the installation

```
python3 -c "import torch, torchvision; print(f'Torch: {torch.__version__}');
print(f'Torchvision: {torchvision.__version__}'); print('CUDA:', torch.cuda.is_available())"
```



References

https://github.com/pytorch/vision

https://docs.ultralytics.com/guides/nvidia-jetson/